

Product Requirements Document (PRD)

OnmiRank — AI Search Visibility Intelligence Platform

Version: 1.0
Date: June 25, 2026
Author: Product Team (Solo Founder + AI Co-Builder)
Status: Ready for Development
Default Language: English (with Indonesian localization support)

1. DOCUMENT CONTROL

1.1 Revision History

Version	Date	Author	Description
1.0	2026-06-25	Product Team	Initial PRD for MVP

1.2 Approval

Role	Name	Date	Signature
Product Owner	[Founder]	TBD	—
Tech Lead	[Founder]	TBD	—

2. EXECUTIVE SUMMARY

2.1 Product Vision

OnmiRank is an **AI Search Visibility Intelligence Platform** that helps brands, solo developers, digital marketers, and content creators become **discoverable and citable** by AI search engines (ChatGPT, Perplexity, Gemini, Kimi, Claude). Unlike traditional SEO tools that focus on Google rankings, OnmiRank optimizes for **Generative Engine Optimization (GEO)** — the new paradigm where AI engines synthesize answers rather than return links.

2.2 Problem Statement

The Shift Nobody Saw Coming:

- **60% of AI searches end without a single click** (zero-click phenomenon)
- Only **16% of brands are visible** in AI-generated answers
- **84% of brands are completely invisible** to AI search engines
- The overlap between Google #1 ranking and AI citation has dropped **below 20%**
- AI traffic converts at **14.2%** — nearly 5x higher than traditional organic traffic

For the Target User:

- Solo developers build apps but remain invisible when users ask AI “what’s the best X tool?”
- Digital marketers manage client brands that don’t appear in ChatGPT/Perplexity answers
- Content creators produce quality content but never get cited by AI engines
- UMKM (Indonesian MSMEs) cannot afford \$399/month enterprise GEO tools

2.3 Solution Overview

OnmiRank provides:

1. **GEO Audit Engine** — Comprehensive scan of technical signals, content structure, and brand entity
2. **Auto-Generate Technical Files** — llms.txt, AI.txt, and Schema.org JSON-LD snippets
3. **AI Citation Monitoring** — Track brand mentions across ChatGPT, Perplexity, Gemini, Kimi, Claude
4. **Competitor GEO Gap Analysis** — Compare your GEO score against competitors
5. **Actionable Reports** — Not just scores, but step-by-step fix instructions

2.4 Success Criteria

Metric	Target (M3)	Target (M6)	Target (M12)
Free Audits Completed	500+	2,000+	5,000+
Email Signups	200+	1,000+	3,000+
Paid Conversions	50+	300+	800+
MRR	\$500	\$2,500	\$10,000+
GEO Score Accuracy	>80%	>85%	>90%
NPS	—	>40	>50

3. MARKET ANALYSIS

3.1 Target Market

Primary Markets:

1. **Solo Developers / Indie Hackers** (Global, English-speaking)
 - Pain: Apps invisible in AI search despite good functionality
 - Willingness to pay: \$9-29/month
 - Channel: Twitter/X, Hacker News, Indie Hackers, Product Hunt
2. **Digital Marketers** (Indonesia + Southeast Asia focus)
 - Pain: Clients asking “why aren’t we in ChatGPT?”
 - Willingness to pay: \$29-99/month
 - Channel: LinkedIn, Facebook Groups, webinars
3. **Content Creators** (Global)
 - Pain: Quality content not cited by AI, stagnant traffic

- Willingness to pay: \$9-29/month
 - Channel: YouTube, TikTok, Threads
4. **UMKM / Small E-commerce** (Indonesia primary)
- Pain: No budget for SEO agencies, need affordable visibility
 - Willingness to pay: \$9/report or Rp 150,000
 - Channel: WhatsApp, Instagram, TikTok Shop
5. **Digital Agencies** (Indonesia + SEA)
- Pain: Need GEO service offering but lack expertise
 - Willingness to pay: \$99-299/month
 - Channel: Direct outreach, partnerships

3.2 Market Size

Segment	TAM	SAM	SOM (Y1)
Global GEO Tools	\$2.1B	\$420M	—
Indonesia Digital Marketing	\$1.8B	\$180M	\$50K
SEA Solo Developers	—	\$85M	\$30K
Total Addressable	—	—	\$80K Y1

3.3 Competitive Analysis

Competitor	Price	Strengths	Weaknesses vs OnmiRank
Profound	\$399+/mo	Enterprise analytics, deep data	No self-serve, US-only, no Indonesia focus
Semrush AI	\$199/mo	Brand recognition, SEO integration	SEO-centric (not GEO-native), expensive for SEA
BrandViz.AI	~\$130/mo	B2B buyer journey tracking	AUD pricing, complex UI, no local context
BotRank.ai	\$89/mo	Multi-engine tracking	Limited action plans, no auto-generate
AI Labs Audit	€79/mo	Simple audit interface	Very limited features, no monitoring
Botric	\$49/mo	GEO suite + AI agents	Shallow for Indonesia, no white-label
Otterly.AI	\$29/mo	Affordable entry point	Complex interface, no white-label, limited data
OnmiRank (Us)	\$9-99/mo	Indonesia-first, auto-generate, affordable	New entrant, building brand awareness

3.4 Competitive Advantage

1. **Indonesia-First Market Position** — Only GEO tool with Indonesian language nuance, local content understanding, and SEA market focus

2. **Auto-Generate Technical Files** — Competitors audit only; OnmiRank generates deployable llms.txt, AI.txt, and Schema JSON-LD
 3. **Per-Unit Pricing** — \$9/report eliminates subscription anxiety; natural upsell path to \$29/month
 4. **Data Moat** — Every audit builds a citation database for Indonesian/SEA queries, improving accuracy over time
 5. **Build-in-Public Distribution** — Founder has existing audience on Threads/TikTok for zero-cost acquisition
-

4. USER PERSONAS

4.1 Persona 1: “DevDan” — Solo Developer

Name: Dan
Age: 28
Location: Jakarta, Indonesia
Role: Solo Developer / Indie Hacker
Income: \$800-1,500/month
Tech Stack: Next.js, Node.js, Prisma

Goals:

- Launch side projects that get discovered
- Build passive income from micro-SaaS
- Minimize marketing effort, maximize reach

Pain Points:

- "I built a great tool but nobody finds it"
- "I don't understand SEO, let alone GEO"
- "I can't afford \$200/month for marketing tools"
- "When people ask ChatGPT for tools like mine, I'm not mentioned"

Behavior:

- Active on Twitter/X and Hacker News
- Builds in public, shares progress
- Values automation and "set it and forget it" solutions
- Willing to pay for tools that save time

OnmiRank Value:

- \$9 audit shows exactly why he's invisible
- Auto-generated files he can deploy in 5 minutes
- Monitor tier (\$29) alerts when his score drops
- "Finally, a marketing tool that speaks developer"

4.2 Persona 2: “MarketerMaya” — Digital Marketer

Name: Maya
Age: 32
Location: Bali, Indonesia
Role: Freelance Digital Marketer
Clients: 8-12 UMKM clients

Income: \$1,500-2,500/month

Goals:

- Offer new services to differentiate from competitors
- Keep clients happy with measurable results
- Scale without hiring more people

Pain Points:

- "Clients keep asking why they're not in ChatGPT"
- "I don't have GEO expertise to sell as a service"
- "Existing tools are too expensive for my clients' budgets"
- "I need white-label reports to look professional"

Behavior:

- Manages client work via WhatsApp and email
- Active in Facebook marketing groups
- Attends webinars and online courses
- Values tools that make her look good to clients

OnmiRank Value:

- White-label GEO reports for all clients
- Competitor gap analysis she can present
- Affordable enough to mark up 2-3x for clients
- "Finally, I can offer GEO without becoming an expert"

4.3 Persona 3: “CreatorCitra” — Content Creator

Name: Citra

Age: 26

Location: Surabaya, Indonesia

Role: Tech Content Creator

Platform: YouTube (50K subs), TikTok (200K followers)

Income: \$1,000-2,000/month (ads + sponsorships)

Goals:

- Grow audience through AI discovery
- Get cited by AI when people ask about her niche
- Monetize expertise through courses/consulting

Pain Points:

- "My old content is buried, never resurfaced by AI"
- "I don't know how to structure content for AI citation"
- "I create content daily but AI never mentions me"
- "SEO tools don't help with AI visibility"

Behavior:

- Creates 3-5 pieces of content per week
- Obsessed with analytics and growth metrics
- Early adopter of new platforms and tools
- Shares discoveries with her audience

OnmiRank Value:

- Shows which of her content is AI-visible
- Content restructuring recommendations
- Freshness alerts: "Update this post, it's losing AI citation"
- "My old content is now working for me again"

5. PRODUCT REQUIREMENTS

5.1 Feature Overview

Feature	Priority	MVP	Release
GEO Audit Engine	P0	Yes	MVP
Auto-Generate llms.txt	P0	Yes	MVP
Auto-Generate AI.txt	P0	Yes	MVP
Auto-Generate Schema	P0	Yes	MVP
JSON-LD			
GEO Score Dashboard	P0	Yes	MVP
PDF Report	P0	Yes	MVP
Generation			
User Authentication	P0	Yes	MVP
Payment Integration	P0	Yes	MVP
AI Citation Monitoring	P1	No	v1.1
Competitor GEO	P1	No	v1.1
Analysis			
White-Label Reports	P1	No	v1.2
API Access	P2	No	v1.3
Team/Agency	P2	No	v1.3
Dashboard			
Chrome Extension	P2	No	v1.4
Slack/Discord	P3	No	v2.0
Integration			

5.2 Detailed Feature Specifications

5.2.1 GEO Audit Engine (P0) User Story:

As a solo developer, I want to enter my website URL and receive a comprehensive GEO visibility score so that I understand why my brand is invisible to AI search engines.

Acceptance Criteria:

1. URL Input
- User can enter any valid URL (with or without protocol)
 - System validates URL format and accessibility
 - Support for subdomains and specific pages
 - Maximum 1 audit per email for free tier
2. Technical Signals Scan

- **robots.txt Analysis:**
 - Detect presence of robots.txt
 - Check if AI crawlers are blocked (GPTBot, ClaudeBot, PerplexityBot, Google-Extended, KimiBot)
 - Score: 0-100 based on crawler accessibility
 - Flag: “Critical — AI crawlers cannot access your site”
- **llms.txt Detection:**
 - Check if llms.txt exists at root domain
 - Parse content quality (completeness, markdown format)
 - Score: 0-100
 - If missing: offer auto-generation
- **AI.txt Detection:**
 - Check if AI.txt exists at root domain
 - Verify purpose-based controls (indexing vs training)
 - Score: 0-100
- **Schema.org Markup:**
 - Detect JSON-LD presence
 - Identify schema types: FAQPage, HowTo, Article, Organization, BreadcrumbList, Product, Review
 - Score per schema type (FAQPage = +20, HowTo = +15, etc.)
 - Total schema score: 0-100
 - Flag missing high-impact schemas

3. Content Structure Analysis

- **Answer-First Detection:**
 - Check if H1/H2 contains direct answer within first 40-60 words
 - Score: 0-100
- **Self-Contained Units (SCUs):**
 - Detect content blocks that can stand alone (paragraph + heading)
 - Score based on SCU density
- **Question-Based Headers:**
 - Count H2/H3 that are phrased as questions
 - Score: percentage of question-based headers
- **Evidence Density:**
 - Check for statistics, data, citations every 150-200 words
 - Flag vague claims without evidence
- **Comparison Tables:**
 - Detect presence of comparison/comparison tables
 - Score bonus for structured comparisons

4. Brand Entity Signals

- **NAP Consistency:**
 - Extract Name, Address, Phone from website
 - Compare against known platforms (Google Business, social media)
 - Score: 0-100 based on consistency
- **Organization Schema:**
 - Check for Organization schema on homepage
 - Verify sameAs properties (LinkedIn, Crunchbase, G2, Capterra)
 - Score: 0-100

- **Third-Party Presence:**
 - Detect mentions on review platforms
 - Check social media profiles
 - Score based on platform diversity

5. Freshness Signals

- **dateModified:** Check if pages have dateModified schema
- **Content Age:** Calculate average age of indexed content
- **Update Frequency:** Detect publishing patterns
- Score: 0-100 (higher for recently updated content)
- Flag: “50% of AI-cited content is <13 weeks old”

6. Overall GEO Score

- Weighted average of all categories:
 - Technical Signals: 25%
 - Content Structure: 30%
 - Brand Entity: 25%
 - Freshness: 20%
- Score range: 0-100
- Grade: F (0-49), D (50-59), C (60-69), B (70-79), A (80-89), A+ (90-100)

Output:

- Interactive dashboard with score breakdown
- Visual radar chart showing strengths/weaknesses
- Prioritized action list (highest impact first)
- Estimated time to implement each fix

5.2.2 Auto-Generate Technical Files (P0) User Story:

As a website owner, I want to automatically generate llms.txt, AI.txt, and Schema JSON-LD snippets so that I can deploy them without writing code.

Acceptance Criteria:

1. Auto-Generate llms.txt

- Parse sitemap.xml or crawl website structure
- Extract page titles, descriptions, and key content
- Generate markdown-formatted llms.txt:

```
# [Brand Name] — AI Content Guide

## Overview
[Brand description in 2-3 sentences]

## Products/Services
- [URL] — [Description]

## Key Content
- [URL] — [Description]
```



```
## Contact
[Contact information]
```

- Allow user editing before download
- One-click download as .txt file
- Deployment instructions included

2. Auto-Generate AI.txt

- Generate purpose-based control file:

```
User-agent: GPTBot
Allow: /
Purpose: indexing

User-agent: ClaudeBot
Allow: /
Purpose: indexing

User-agent: PerplexityBot
Allow: /
Purpose: indexing

User-agent: Google-Extended
Allow: /
Purpose: indexing
```

- Options: “Allow All”, “Indexing Only”, “Block Training”
- Custom user-agent support
- One-click download

3. Auto-Generate Schema JSON-LD

- Based on detected content type, generate appropriate schema:
 - FAQPage schema for FAQ sections
 - HowTo schema for tutorial content
 - Organization schema for homepage
 - Article schema for blog posts
 - Product schema for e-commerce
- Validate against Schema.org standards
- Copy-to-clipboard functionality
- Implementation instructions (where to paste in HTML)

5.2.3 GEO Score Dashboard (P0) User Story:

As a user, I want a visual dashboard showing my GEO score history and trends so that I can track improvement over time.

Acceptance Criteria:

1. Score Overview

- Large, prominent GEO score display (0-100)
- Color-coded: Red (<50), Orange (50-69), Yellow (70-79), Green (80-89), Dark Green (90-100)
- Grade letter displayed prominently

- “You’re in the top X% of websites” comparison

2. Category Breakdown

- Expandable sections for each category:
 - Technical Signals (0-100)
 - Content Structure (0-100)
 - Brand Entity (0-100)
 - Freshness (0-100)
- Each section shows:
 - Current score
 - Previous score (if available)
 - Trend arrow (up/down/stable)
 - Specific issues found
 - Recommended fixes

3. Historical Tracking

- Line chart showing score over time
- Filter by: 7 days, 30 days, 90 days, All time
- Annotations for major updates
- Export data as CSV

4. Action Items

- Prioritized to-do list
- Each item shows:
 - Impact (High/Medium/Low)
 - Effort (Quick/Medium/Complex)
 - Estimated score improvement
 - Link to relevant tool/resource
- Check off completed items
- Progress bar: “X of Y fixes completed”

5.2.4 PDF Report Generation (P0) User Story:

As a digital marketer, I want to download a professional PDF report of my GEO audit so that I can share it with clients or stakeholders.

Acceptance Criteria:

1. Report Content

- Cover page: Brand name, URL, audit date, overall score
- Executive summary (1 page)
- Detailed breakdown per category
- Visual charts and graphs
- Prioritized action plan
- Technical appendix (detected schemas, robots.txt content, etc.)

2. Design

- Professional, clean layout
- Brand colors (OnmiRank branding)
- Mobile-friendly if viewed on device
- Print-optimized

3. White-Label (Paid Tier)

- Remove OnmiRank branding
- Add user's logo and colors
- Custom cover page
- "Generated by [User Agency Name]" footer

4. Delivery

- One-click PDF generation
 - Download link (valid 30 days)
 - Email delivery option
-

5.2.5 User Authentication (P0) User Story:

As a user, I want to create an account and log in so that I can save my audits and track progress over time.

Acceptance Criteria:

1. Registration

- Email + password registration
- OAuth: Google, GitHub
- Email verification required
- Password requirements: min 8 chars, 1 uppercase, 1 number

2. Login

- Email + password
- OAuth login
- "Remember me" option
- Password reset via email

3. Account Management

- Profile settings (name, company, timezone)
- Notification preferences
- Billing information
- Subscription management

4. Free Tier Limitations

- 1 audit per email address
 - Basic report only (no PDF)
 - No historical tracking
 - No monitoring
-

5.2.6 Payment Integration (P0) User Story:

As a user, I want to purchase audit credits or subscribe to a plan so that I can access premium features.

Acceptance Criteria:

1. Pricing Tiers

- **Free:** \$0 — 1 audit, basic report
- **Pay-Per-Audit:** \$9/report — full audit + PDF + auto-generate files

- **Monitor:** \$29/month — 10 audits/month + monitoring + alerts + historical tracking
- **Agency:** \$99/month — 50 audits/month + white-label + competitor analysis + 5 team members
- **Enterprise:** Custom pricing — unlimited + API + dedicated support

2. Payment Methods

- Credit/Debit card (Stripe)
- Indonesian local payment (Midtrans: GoPay, OVO, DANA, bank transfer, QRIS)
- PayPal (international)

3. Billing

- Monthly subscription with auto-renew
- One-time purchase for audit credits
- Invoice generation
- Prorated upgrades/downgrades

4. Trial

- 7-day free trial for Monitor tier
- No credit card required for trial
- Convert to paid or downgrade to free

5.2.7 AI Citation Monitoring (P1 — v1.1) User Story:

As a user, I want to know when my brand is mentioned or cited by AI search engines so that I can track my GEO progress.

Acceptance Criteria:

1. Citation Tracking

- Monitor across: ChatGPT, Perplexity, Gemini, Kimi, Claude
- Detect brand name mentions in AI responses
- Track URL citations
- Score: “Your brand was cited X times this week”

2. Query Monitoring

- Track specific keywords/queries
- “When people ask about [topic], is your brand mentioned?”
- Weekly report of tracked queries

3. Alerts

- Email notification when brand is newly cited
- Alert when citation count drops
- Alert when competitor gets cited for your target queries

4. Dashboard

- Citation trend chart
- Top queries where brand appears
- Comparison vs previous period

5.2.8 Competitor GEO Analysis (P1 — v1.1) User Story:

As a digital marketer, I want to compare my GEO score against my competitors so that I can identify gaps and opportunities.

Acceptance Criteria:

1. Competitor Addition

- Add up to 3 competitors (Monitor tier) or 10 (Agency tier)
- Enter competitor URLs
- System runs GEO audit on each competitor

2. Comparison Dashboard

- Side-by-side score comparison
- Radar chart: You vs Competitor A vs Competitor B
- Gap analysis: “Competitor X has FAQPage schema, you don’t”
- Opportunity identification

3. Benchmarking

- Industry average GEO score
 - Percentile ranking
 - “You’re in the top 15% of [industry] websites”
-

5.2.9 White-Label Reports (P1 — v1.2) User Story:

As an agency owner, I want to generate GEO reports with my own branding so that I can present them to clients as my own service.

Acceptance Criteria:

1. Branding Customization

- Upload agency logo
- Set primary/secondary colors
- Custom cover page template
- Custom footer text
- Custom domain for report links (CNAME)

2. Client Management

- Add client profiles
- Assign audits to clients
- Client-specific dashboards
- Client invitation system

3. Report Templates

- Multiple report templates
 - Executive summary vs detailed technical
 - Scheduled report generation (weekly/monthly)
-

5.3 Non-Functional Requirements

5.3.1 Performance

Requirement	Target	Measurement
Audit completion time	<30 seconds	95th percentile
Dashboard load time	<2 seconds	95th percentile
PDF generation	<10 seconds	95th percentile
Concurrent audits	50+	Load testing
Uptime	99.9%	Monitoring

5.3.2 Scalability

- Support 10,000+ monthly active users by M12
- Handle 1,000+ concurrent audits
- Database: horizontal scaling ready
- CDN for static assets

5.3.3 Security

- HTTPS everywhere
- API rate limiting (100 requests/minute per IP)
- Input sanitization and validation
- SQL injection prevention
- XSS protection
- CSRF tokens for forms
- GDPR compliance (data deletion, export)
- Secure payment processing (PCI DSS via Stripe)

5.3.4 Accessibility

- WCAG 2.1 Level AA compliance
- Keyboard navigation support
- Screen reader compatibility
- Color contrast ratios
- Alt text for all images

5.3.5 Internationalization

- Default language: English
- Indonesian (Bahasa Indonesia) support
- RTL language support (future)
- Currency: USD (default), IDR (Indonesia)
- Date/time formatting per locale

6. USER INTERFACE / USER EXPERIENCE

6.1 Information Architecture

OnmiRank
└─ Landing Page
└─ Hero Section (value prop + free audit CTA)

```

|   └─ How It Works (3 steps)
|   └─ Features Overview
|   └─ Pricing
|   └─ Testimonials
|   └─ FAQ
|
| └─ Dashboard (Authenticated)
|   └─ Overview (GEO Score + trends)
|   └─ Audits
|     └─ New Audit
|     └─ Audit History
|     └─ Audit Detail
|   └─ Reports
|     └─ Generated Reports
|     └─ Download Center
|   └─ Monitoring (Paid)
|     └─ Citation Tracker
|     └─ Query Monitor
|     └─ Alerts
|   └─ Competitors (Paid)
|     └─ Competitor List
|     └─ Comparison
|   └─ Tools
|     └─ llms.txt Generator
|     └─ AI.txt Generator
|     └─ Schema Generator
|   └─ Settings
|     └─ Profile
|     └─ Billing
|     └─ Notifications
|     └─ Team (Agency)
|
| └─ Free Audit (No Auth Required)
|   └─ Single Page Audit → Email Results
|
└─ Auth
  └─ Login
  └─ Register
  └─ Forgot Password
  └─ OAuth Callback

```

6.2 Key Screens

6.2.1 Landing Page Hero Section:

- **Headline:** “Is Your Brand Invisible to AI Search?”
- **Subheadline:** “84% of brands never appear in ChatGPT, Perplexity, or Gemini. OnmiRank fixes that.”
- **CTA:** “Get Your Free GEO Score” (URL input field)
- **Visual:** Animated radar chart showing GEO score

Social Proof:

- “Join 500+ brands optimizing for AI search”
- Logos of early adopters (when available)
- Testimonial carousel

How It Works:

1. Enter your URL → 2. Get GEO Score → 3. Fix & Monitor
- Each step with icon and 1-sentence description

Pricing Section:

- 4-tier comparison table
- “Most Popular” badge on Monitor tier
- FAQ accordion below

6.2.2 Audit Results Page Header:

- URL audited
- Audit date
- Overall GEO Score (large, color-coded)
- Grade badge (A+, A, B, C, D, F)
- “You’re in the top X%” comparison

Score Breakdown:

- 4 expandable cards (Technical, Content, Brand, Freshness)
- Each card: score, trend, key issues, fix button
- Radar chart in center

Action Items:

- Prioritized list
- Each item: impact level, effort level, estimated score gain
- “Fix Now” button → opens relevant tool

Tools Section:

- llms.txt generator
- AI.txt generator
- Schema snippets
- Download buttons

CTA:

- “Want to track progress? Upgrade to Monitor (\$29/month)”
- “Download PDF Report” (gated for free tier)

6.2.3 Dashboard Overview Tab:

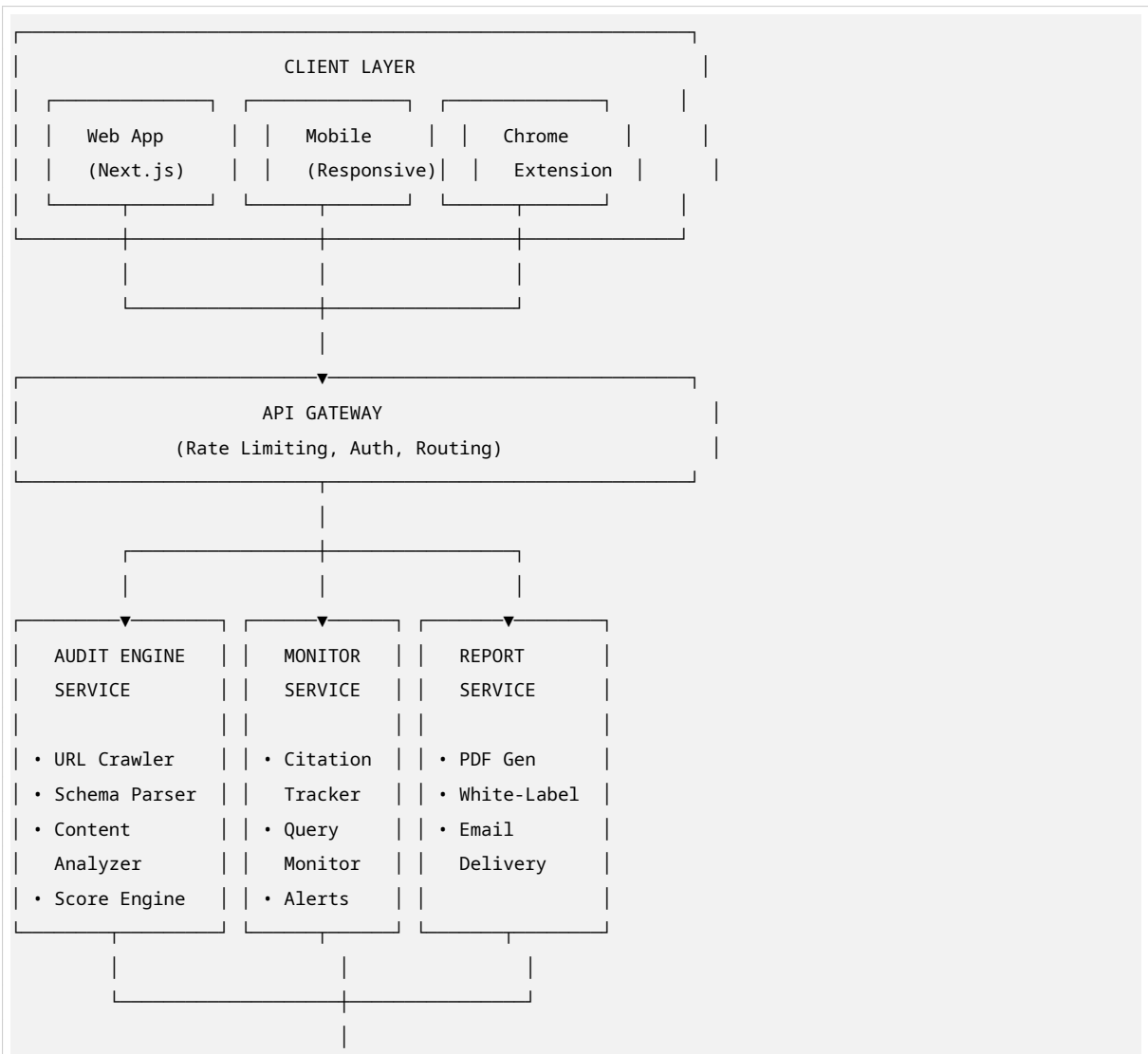
- GEO Score trend chart (line chart, 30/90/All time)
- Recent audits list
- Active alerts/notifications
- Quick actions: “New Audit”, “Generate Report”, “Add Competitor”

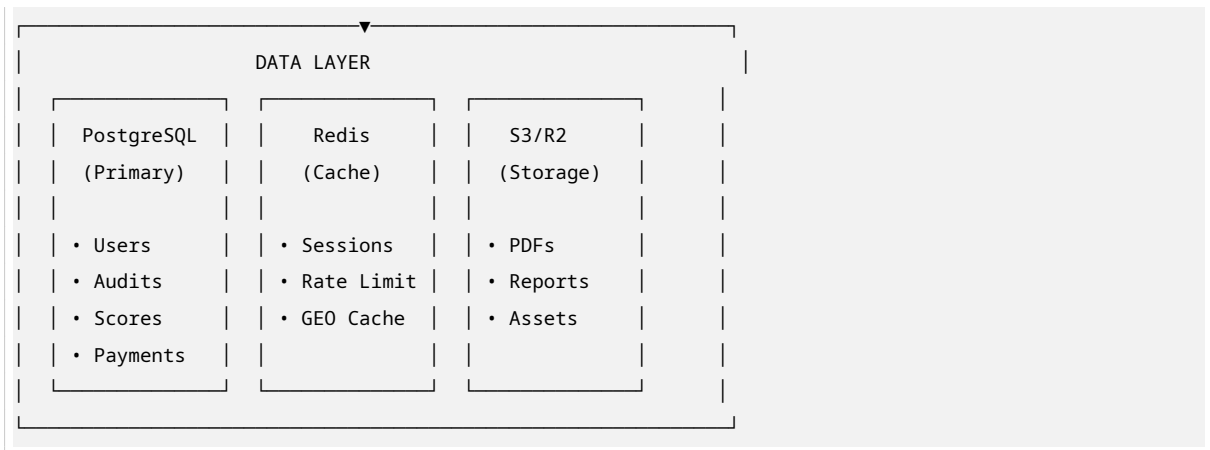
Monitoring Tab (Paid):

- Citation count this week/month
- Top queries chart
- Recent citations list
- Competitor citation comparison

7. TECHNICAL ARCHITECTURE

7.1 System Architecture





7.2 Technology Stack

Layer	Technology	Justification
Frontend	Next.js 15 (App Router)	SSR for SEO, familiar stack, Vercel optimization
UI Framework	shadcn/ui + Tailwind CSS	Consistent design system, rapid development
Backend	Next.js API Routes + tRPC	Full-stack TypeScript, type-safe APIs
Database	PostgreSQL (Neon)	Serverless Postgres, cost-effective, scalable
ORM	Prisma	Type-safe database queries, migration management
Cache	Redis (Upstash)	Session management, rate limiting, GEO score caching
File Storage	Cloudflare R2	S3-compatible, zero egress fees, global CDN
Web Crawler	Puppeteer + Cheerio	Headless browser for JS-rendered content, HTML parsing
AI/LLM	OpenAI GPT-4o / Kimi API	Report generation, content analysis, recommendations
PDF Generation	Puppeteer + HTML template	Server-side PDF rendering
Auth	NextAuth.js v5	OAuth (Google, GitHub), JWT sessions
Payments	Stripe + Midtrans	International + Indonesia local payments
Email	Resend	Transactional emails, report delivery
Analytics	PostHog	Product analytics, funnel tracking, session replay
Monitoring	Vercel Analytics + Sentry	Performance monitoring, error tracking
Hosting	Vercel	Edge network, automatic scaling, familiar platform
Domain	Cloudflare	DNS, SSL, DDoS protection

7.3 Database Schema

```
-- Users Table
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email VARCHAR(255) UNIQUE NOT NULL,
  name VARCHAR(255),
```

```

avatar_url TEXT,
company VARCHAR(255),
timezone VARCHAR(50) DEFAULT 'Asia/Jakarta',
locale VARCHAR(10) DEFAULT 'en',
currency VARCHAR(3) DEFAULT 'USD',
subscription_tier VARCHAR(20) DEFAULT 'free', -- free, pay_per_audit, monitor, agency, enterprise
subscription_status VARCHAR(20) DEFAULT 'inactive', -- active, trialing, canceled, past_due
stripe_customer_id VARCHAR(255),
midtrans_customer_id VARCHAR(255),
credits_remaining INT DEFAULT 0, -- for pay-per-audit
audits_used_this_month INT DEFAULT 0,
max_audits_per_month INT DEFAULT 1,
white_label_enabled BOOLEAN DEFAULT FALSE,
white_label_logo_url TEXT,
white_label_primary_color VARCHAR(7) DEFAULT '#0f172a',
white_label_secondary_color VARCHAR(7) DEFAULT '#3b82f6',
created_at TIMESTAMP DEFAULT NOW(),
updated_at TIMESTAMP DEFAULT NOW()
);

-- Audits Table
CREATE TABLE audits (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,
  url TEXT NOT NULL,
  domain VARCHAR(255) NOT NULL,
  overall_score INT CHECK (overall_score >= 0 AND overall_score <= 100),
  grade VARCHAR(2), -- A+, A, B, C, D, F
  technical_score INT,
  content_score INT,
  brand_score INT,
  freshness_score INT,
  status VARCHAR(20) DEFAULT 'pending', -- pending, processing, completed, failed
  report_url TEXT,
  pdf_url TEXT,
  is_white_label BOOLEAN DEFAULT FALSE,
  client_id UUID, -- for agency tier
  created_at TIMESTAMP DEFAULT NOW(),
  completed_at TIMESTAMP
);

-- Audit Details Table
CREATE TABLE audit_details (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  audit_id UUID REFERENCES audits(id) ON DELETE CASCADE,
  category VARCHAR(50) NOT NULL, -- technical, content, brand, freshness
  subcategory VARCHAR(100) NOT NULL,
  score INT CHECK (score >= 0 AND score <= 100),
  max_score INT,
  findings JSONB, -- structured findings
  recommendations JSONB, -- actionable recommendations

```

```

        created_at TIMESTAMP DEFAULT NOW()
    );

-- Action Items Table
CREATE TABLE action_items (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    audit_id UUID REFERENCES audits(id) ON DELETE CASCADE,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    category VARCHAR(50),
    impact VARCHAR(20), -- high, medium, low
    effort VARCHAR(20), -- quick, medium, complex
    estimated_score_gain INT,
    is_completed BOOLEAN DEFAULT FALSE,
    completed_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT NOW()
);

-- Competitors Table
CREATE TABLE competitors (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    name VARCHAR(255) NOT NULL,
    url TEXT NOT NULL,
    domain VARCHAR(255) NOT NULL,
    last_audit_id UUID REFERENCES audits(id),
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT NOW()
);

-- Citation Tracking Table
CREATE TABLE citations (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    domain VARCHAR(255) NOT NULL,
    ai_engine VARCHAR(50) NOT NULL, -- chatgpt, perplexity, gemini, kimi, claude
    query TEXT NOT NULL,
    response_snippet TEXT,
    is_cited BOOLEAN DEFAULT FALSE,
    citation_url TEXT,
    citation_context TEXT,
    tracked_at TIMESTAMP DEFAULT NOW()
);

-- Tracked Queries Table
CREATE TABLE tracked_queries (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    query TEXT NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    last_checked_at TIMESTAMP,

```

```

        created_at TIMESTAMP DEFAULT NOW()
    );

-- Generated Files Table
CREATE TABLE generated_files (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    audit_id UUID REFERENCES audits(id),
    file_type VARCHAR(50) NOT NULL, -- llms_txt, ai_txt, schema_faq, schema_howto, etc.
    content TEXT NOT NULL,
    download_url TEXT,
    created_at TIMESTAMP DEFAULT NOW()
);

-- Payments Table
CREATE TABLE payments (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    amount DECIMAL(10,2) NOT NULL,
    currency VARCHAR(3) DEFAULT 'USD',
    payment_method VARCHAR(50), -- stripe, midtrans, paypal
    payment_intent_id VARCHAR(255),
    status VARCHAR(20), -- pending, completed, failed, refunded
    product_type VARCHAR(50), -- subscription, audit_credit, service
    description TEXT,
    created_at TIMESTAMP DEFAULT NOW()
);

-- Clients Table (Agency Tier)
CREATE TABLE clients (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    agency_user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    name VARCHAR(255) NOT NULL,
    email VARCHAR(255),
    logo_url TEXT,
    primary_color VARCHAR(7) DEFAULT '#0f172a',
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT NOW()
);

-- Alerts Table
CREATE TABLE alerts (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID REFERENCES users(id) ON DELETE CASCADE,
    type VARCHAR(50) NOT NULL, -- score_drop, new_citation, competitor_cited, freshness_warning
    title VARCHAR(255) NOT NULL,
    message TEXT,
    is_read BOOLEAN DEFAULT FALSE,
    data JSONB,
    created_at TIMESTAMP DEFAULT NOW()
);

```

```
-- Indexes
CREATE INDEX idx_audits_user_id ON audits(user_id);
CREATE INDEX idx_audits_domain ON audits(domain);
CREATE INDEX idx_audits_created_at ON audits(created_at);
CREATE INDEX idx_citations_user_id ON citations(user_id);
CREATE INDEX idx_citations_domain ON citations(domain);
CREATE INDEX idx_citations_tracked_at ON citations(tracked_at);
CREATE INDEX idx_action_items_audit_id ON action_items(audit_id);
CREATE INDEX idx_payments_user_id ON payments(user_id);
```

7.4 API Endpoints

Authentication:

```
POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout
POST /api/auth/forgot-password
POST /api/auth/reset-password
GET /api/auth/session
```

Audit:

```
POST /api/audits          — Create new audit (free tier limited)
GET /api/audits           — List user's audits
GET /api/audits/:id       — Get audit details
GET /api/audits/:id/report — Get full report
GET /api/audits/:id/pdf   — Generate/download PDF
DELETE /api/audits/:id    — Delete audit
```

Tools:

```
POST /api/tools/llms-txt   — Generate llms.txt
POST /api/tools/ai-txt     — Generate AI.txt
POST /api/tools/schema     — Generate Schema JSON-LD
GET /api/tools/schema/types — List available schema types
```

Monitoring:

```
GET /api/monitoring/citations — Get citation data
GET /api/monitoring/queries   — List tracked queries
POST /api/monitoring/queries  — Add tracked query
DELETE /api/monitoring/queries/:id — Remove tracked query
GET /api/monitoring/alerts    — Get alerts
PATCH /api/monitoring/alerts/:id — Mark alert as read
```

Competitors:

```
GET /api/competitors       — List competitors
POST /api/competitors      — Add competitor
DELETE /api/competitors/:id — Remove competitor
GET /api/competitors/compare — Compare scores
```

Billing:

```
GET /api/billing/subscription — Get current subscription
```

```
POST /api/billing/checkout      — Create checkout session
POST /api/billing/portal       — Open billing portal
GET  /api/billing/invoices     — List invoices
POST /api/billing/credits      — Purchase audit credits
```

User:

```
GET  /api/user/profile         — Get profile
PATCH /api/user/profile       — Update profile
GET  /api/user/usage           — Get usage stats
GET  /api/user/settings        — Get settings
PATCH /api/user/settings      — Update settings
```

Admin (Future):

```
GET  /api/admin/users
GET  /api/admin/audits
GET  /api/admin/metrics
```

8. DATA FLOW

8.1 Audit Flow

```
User enters URL
|
▼
[Validation] → Is valid URL? → No → Return error
| Yes
▼
[Rate Limit Check] → Exceeded? → Yes → Return 429
| No
▼
[Free Tier Check] → Already used free audit? → Yes → Prompt upgrade
| No
▼
[Crawl Website]
├─ Fetch robots.txt
├─ Fetch llms.txt
├─ Fetch AI.txt
├─ Fetch sitemap.xml
├─ Crawl homepage + key pages (max 10)
├─ Extract HTML content
|
▼
[Analysis Engine]
├─ Parse schemas (JSON-LD, microdata)
├─ Analyze content structure
├─ Check brand entity signals
├─ Evaluate freshness
├─ Calculate scores
|
▼
```

```
[AI Report Generation]
├─ Generate findings summary
├─ Create prioritized action items
├─ Generate recommendations
└─ Calculate overall GEO Score
|
▼

[Store Results]
├─ Save audit to database
├─ Cache results (Redis, 24h)
└─ Generate PDF (async)
|
▼

[Return Response]
├─ Audit ID
├─ GEO Score
├─ Category breakdowns
├─ Action items
└─ Generated files (if applicable)
```

8.2 Citation Monitoring Flow (v1.1)

```
Scheduled Job (Daily/Weekly)
|
▼

[Fetch Tracked Queries]
├─ For each user
└─ For each tracked query
|
▼

[Query AI Engines]
├─ ChatGPT API (or web scraping)
├─ Perplexity API
├─ Gemini API
└─ Kimi API (future)
|
▼

[Parse Responses]
├─ Extract mentions of user's domain/brand
├─ Extract cited URLs
└─ Extract context snippets
|
▼

[Compare with Previous]
├─ New citation? → Alert user
├─ Citation lost? → Alert user
└─ Score change? → Alert user
|
▼

[Store & Notify]
└─ Save to citations table
```


- └─ Create alert record
- └─ Send email notification (if enabled)

9. SECURITY & COMPLIANCE

9.1 Authentication & Authorization

- **JWT-based sessions** with HttpOnly cookies
- **OAuth 2.0** for Google and GitHub login
- **Role-based access control (RBAC):**
 - **free:** 1 audit, basic report
 - **pay_per_audit:** unlimited audits, full reports, auto-generate
 - **monitor:** + monitoring, alerts, historical tracking, 3 competitors
 - **agency:** + white-label, 10 competitors, 5 team members, 50 audits/month
 - **enterprise:** custom limits, API access, dedicated support
- **API key authentication** for future API access (Enterprise tier)

9.2 Data Protection

- **Encryption at rest:** PostgreSQL with AES-256
- **Encryption in transit:** TLS 1.3
- **PII handling:** Minimal collection (email, name, company optional)
- **Data retention:** Audit data retained for 2 years, user can request deletion
- **GDPR compliance:**
 - Right to access: Export all user data
 - Right to erasure: Delete account + all data
 - Right to portability: JSON export
- **Indonesia data protection:** Comply with UU PDP (when enacted)

9.3 Payment Security

- **PCI DSS compliance** via Stripe (never touch card data)
- **Midtrans** for Indonesia (PCI DSS Level 1)
- **Webhook verification** with signature validation
- **Idempotency keys** for payment operations

9.4 Crawler Ethics

- **Respect robots.txt:** Do not crawl if Disallow
- **Rate limiting:** Max 1 request per second per domain
- **User-Agent:** Identify as “OnmiRankBot/1.0 (+https://onmirank.com/bot)”
- **Cache aggressively:** Minimize repeat crawls (24h cache)
- **No content storage:** Store only analysis results, not full page content

10. TESTING STRATEGY

10.1 Testing Levels

Level	Type	Coverage Target	Tools
Unit	Function/component	>80%	Vitest, React Testing Library
Integration	API endpoints	>70%	Vitest + MSW
E2E	Critical user flows	100%	Playwright
Performance	Load, stress	Key flows	k6, Lighthouse
Security	OWASP Top 10	All	OWASP ZAP, manual
Accessibility	WCAG 2.1 AA	All	axe-core, Lighthouse

10.2 Critical Test Scenarios

1. Audit Flow
 - Valid URL → Complete audit in <30s
 - Invalid URL → Clear error message
 - Blocked robots.txt → Flag in report
 - Missing schema → Recommendations generated
2. Payment Flow
 - Free tier: 1 audit limit enforced
 - Pay-per-audit: Credit deducted, report delivered
 - Subscription: Upgrade/downgrade proration
 - Failed payment: Grace period, retry logic
3. Auth Flow
 - Registration with email verification
 - OAuth login (Google, GitHub)
 - Password reset
 - Session expiration handling
4. PDF Generation
 - Report generates in <10s
 - White-label branding applied correctly
 - Large reports (>50 pages) don't timeout

10.3 Test Data

- Mock websites:** Create 10 test domains with varying GEO scores
- Edge cases:** Empty pages, JS-only sites, blocked crawlers
- Competitor data:** 5 known competitors for comparison testing

11. DEPLOYMENT PLAN

11.1 Environment Strategy

Environment	Purpose	URL	Auto-deploy
Local	Development	localhost:3000	Manual
Preview	PR testing	*.vercel.app	On PR
Staging	Pre-production	staging.onmيران.com	On merge to develop

Table 10 – continued

Environment	Purpose	URL	Auto-deploy
Production	Live	onmirank.com	On merge to main

11.2 MVP Launch Checklist

Week 1-2: Core Development

- Project setup (Next.js, Prisma, shadcn)
- Database schema + migrations
- Auth system (NextAuth + OAuth)
- Audit engine (crawler + analyzer)
- Score calculation logic
- Report generation (HTML + PDF)

Week 3: Tools & Polish

- llms.txt generator
- AI.txt generator
- Schema JSON-LD generator
- Landing page
- Dashboard UI
- Payment integration (Stripe)

Week 4: Testing & Launch

- Unit tests (>80%)
- E2E tests (critical flows)
- Performance optimization
- Security audit
- Production deployment
- Domain setup (onmirank.com)
- SSL certificate
- Analytics setup (PostHog)
- Launch announcement (Threads, TikTok, Product Hunt)

11.3 Post-Launch (Month 2-3)

- Monitor error rates (Sentry)
- Track conversion funnels (PostHog)
- Gather user feedback
- Fix critical bugs

- Optimize audit speed
- Add Midtrans (Indonesia payments)
- Begin v1.1 development (Monitoring)

12. RISKS & MITIGATION

Risk	Probability	Impact	Mitigation
llms.txt not adopted by AI providers	Medium	High	Focus on proven signals (schema, content structure) as primary value
Competitor (Semrush) launches GEO Indonesia	Medium	High	Speed to market, community moat, Indonesia nuance
AI search engines change algorithms frequently	High	Medium	Agile scoring model, weekly updates, transparent methodology
Low free-to-paid conversion	Medium	High	Strong email nurture, clear value prop, LTD for early adopters
Citation tracking technically complex	Medium	High	Start with simulated/manual, automate gradually, partner with AI providers
Server costs exceed revenue early	Medium	High	Aggressive caching, rate limiting, credit system for heavy users
Data privacy concerns (crawling websites)	Low	High	Clear crawler ethics, respect robots.txt, no content storage, GDPR compliance
Founder burnout (solo dev)	Medium	High	Scope MVP tightly, automate where possible, consider co-founder

13. METRICS & ANALYTICS

13.1 North Star Metric

Monthly AI Visibility Score Improvement — Average GEO score increase for active users month-over-month.

13.2 Key Metrics

Category	Metric	Target (M3)	Target (M6)	Target (M12)
Acquisition	Website visitors	5,000/mo	15,000/mo	50,000/mo
	Free audits	500	2,000	5,000
	Email signups	200	1,000	3,000
Activation	Audit completion rate	>70%	>75%	>80%
	Score viewed	>90%	>90%	>90%
	MRR	\$500	\$2,500	\$10,000
Revenue	Paid conversions	50	300	800
	ARPU	\$15	\$20	\$25
	LTV	\$50	\$100	\$200
Retention	Monthly active users	200	800	2,500
	Churn rate	<10%	<8%	<5%
	NPS	—	>40	>50
Engagement	Avg audits per user	2	3	4
	Tool usage rate	30%	40%	50%
	Report downloads	100	500	2,000

13.3 Analytics Implementation

- **PostHog:** Product analytics, funnel tracking, session replay
- **Vercel Analytics:** Web performance, Core Web Vitals
- **Sentry:** Error tracking, performance monitoring
- **Stripe Dashboard:** Revenue metrics, churn analysis
- **Custom dashboard:** GEO score trends, citation data

14. FUTURE ROADMAP

14.1 Version Timeline

Version	Timeline	Key Features
MVP	Month 1-2	Audit engine, auto-generate files, basic dashboard, payments
v1.1	Month 3-4	AI citation monitoring, query tracking, alerts
v1.2	Month 4-5	Competitor GEO analysis, white-label reports

Table 13 – continued

Version	Timeline	Key Features
v1.3	Month 6-7	API access, team/agency dashboard, webhooks
v1.4	Month 7-8	Chrome extension, Slack/Discord integration
v2.0	Month 9-12	AI content optimizer, automated GEO fixes, marketplace

14.2 Long-Term Vision (Year 2-3)

1. **AI Content Optimizer:** Automatically rewrite content for GEO using AI
2. **Automated GEO Fixes:** One-click deployment of fixes (schema injection, content updates)
3. **GEO Marketplace:** Buy/sell GEO-optimized content templates
4. **Enterprise Features:** SSO, SAML, custom AI model training
5. **Global Expansion:** Support for Thai, Vietnamese, Filipino markets
6. **Mobile App:** iOS/Android for monitoring on-the-go

15. APPENDICES

15.1 Glossary

Term	Definition
GEO	Generative Engine Optimization —optimizing content for AI search engines
AI Search	Search engines that use LLMs to synthesize answers (ChatGPT, Perplexity, etc.)
llms.txt	Proposed standard file to guide LLMs on which content to index
AI.txt	Purpose-based control file for AI crawlers (indexing vs training)
Schema.org	Structured data vocabulary for search engines
JSON-LD	JavaScript Object Notation for Linked Data —format for schema markup
SCU	Self-Contained Unit —content block that can stand alone
NAP	Name, Address, Phone —consistency signals for brand entity
E-E-A-T	Experience, Expertise, Authoritativeness, Trustworthiness
Zero-Click	Search that ends without user clicking any result
MRR	Monthly Recurring Revenue
ARPU	Average Revenue Per User
LTV	Lifetime Value
NPS	Net Promoter Score

15.2 Reference Links

- **GEO Research:** Profound GEO 2026 Report, BrightEdge GEO Guide
- **llms.txt Standard:** <https://llmstxt.org/>

- **AI.txt Standard:** <https://ai.txt/>
- **Schema.org:** <https://schema.org/>
- **Competitors:** Profound.ai, Semrush AI, BrandViz.AI, BotRank.ai

15.3 Open Questions

1. Should we support crawling SPAs (Single Page Applications) with heavy JavaScript?
2. What's the pricing for Enterprise tier? Custom per-quote?
3. Should we build a referral/affiliate program?
4. Do we need a mobile app, or is responsive web sufficient?
5. Should we offer GEO implementation as a managed service?

End of PRD

Document Owner: [Founder Name]

Last Updated: June 25, 2026

Next Review: July 25, 2026 (post-MVP launch)